



My warmest and sincerest thanks to all those involved with the production of this most horrendous burden. I must also thank one very special business manager for being especially tolerant of my hyper-active, moody, and insulting nature. (Though I'm not saying that that person didn't deserve most of it.) Mr. Tony Wong deserves special recognition for allowing us to pull an all-nighter at his house so that we could finalize all changes of this magazine. I also have to thank two wonderful and concerned people, Dr. Rothenberg and Mr. Clancy, for nagging to me about the magazine until the whole nuisance was over. All the math teachers also deserve a special thanks for persuading their students to buy our magazine year after year.

This is the very first Math Survey of the 90's and I hope that future editors and staff members will enjoy the success that this staff and I are feeling right now!

A handwritten signature in cursive script, appearing to read "Alfred Lin".

We would like to dedicate this magazine to Mr. Phillips who allowed us to constantly badger him about all sorts of idiotic notions. We thank him for his understanding, cooperation, and patience. We are also indebted to him for his help in countless occasions when we were too busy or too confused to even ask for help.

# Math Survey

The Mathematics Publication of Stuyvesant High School

Volume **LXXIV**  
Spring 1990

*In questions of mathematics the authority  
of a thousand is not worth the humble  
reasoning of a single individual.*

—Galileo Galilei

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This magazine is an attempt to convey the  
diversity, the unity, the beauty, and the joy  
of the mathematical world to the high school community.

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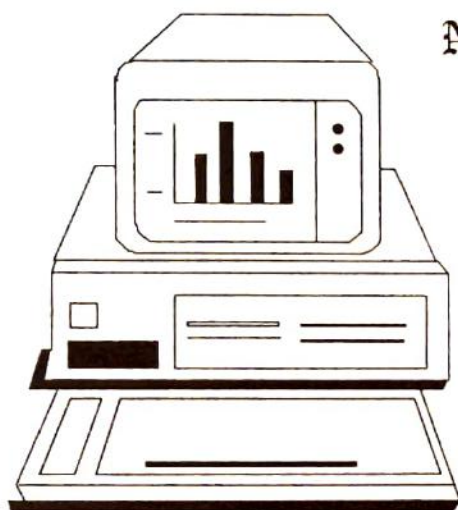
**for the LXIV Edition of  
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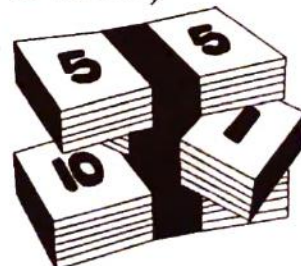
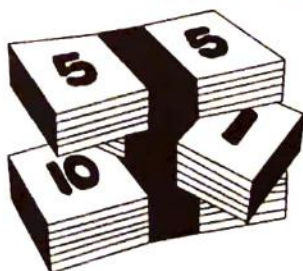
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# Partitions and the Ballot Problem

by Tony Wong

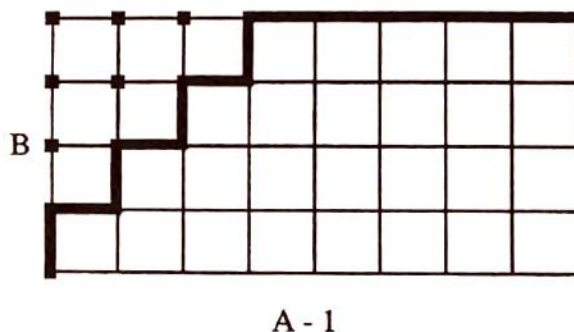
The ballot problem is one of the most famous problems in combinatorial theory, and interestingly enough, it has applications in another field of combinatorics, the theory of partitions. Briefly, the weak-lead ballot problem may be stated as follows: two candidates are running for an office, and the election has been completed. As the votes are counted, it is found that the eventual winner (candidate A) always has at least as many votes as the loser (candidate B). What is the probability of this occurring?

The most straightforward method of solving this problem requires that a rectangular grid be drawn so that each unit in the x-direction is a vote for candidate A and each unit in the y-direction is a vote for candidate B. As the votes are counted, a zig-zag path, or line of route, is traced from the lower left corner to the upper right corner, according to the sequence of votes. A point which goes above the line  $y = x$  violates the condition that candidate B never leads.

If candidate A receives A votes and candidate B receives B votes, then the total number of possible paths across the grid is the number of permutations of the A votes together with the B votes, where votes of the same kind are identical; hence:

$$\frac{(A+B)!}{A! B!}$$

To find the number of paths which do not pass through a point whose ordinate is greater than its abscissa, we can drop the first column of the grid, since from (0,0) we can only go to (1,0). There remain  $[(A+B-1)!] \div [(A-1)! B!]$  possible paths, but those which pass through the points marked below cannot be counted. (The path shown represents the most favorable counting of votes possible for candidate B.)



When the number of these paths is subtracted, the result is:

$$\begin{aligned} \frac{(A+B-1)!}{(A-1)! B!} - \frac{(A+B-1)!}{(A+1)! (B-2)!} &= \frac{(A^2+A)(A+B-1)! - (B^2-B)(A+B-1)!}{(A+1)! B!} \\ &= \frac{(A-B+1)(A+B)!}{(A+1)! B!} \end{aligned}$$

To find the probability of selecting a proper path, therefore, we divide this by the original number of paths:

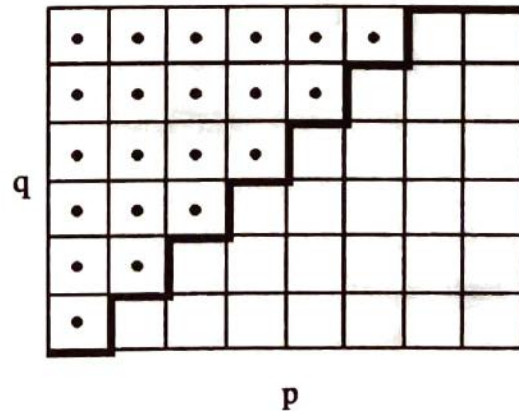
$$\frac{(A-B+1)(A+B)!}{(A+1)! B!} \div \frac{(A+B)!}{A! B!} = \frac{A-B+1}{A+1}$$

Similarly, the probability of selecting an unsatisfactory path is:

$$1 - \frac{A-B+1}{A+1} = \frac{B}{A+1}$$

### Generalization of Integer Partitions

The solution of the ballot problem concerned itself with finding the probability that a randomly selected lattice path would never pass above the line  $y = x$ . There is another way of interpreting this, as can be seen from the diagram below. The line of route separates the lattice into two regions, and the one on the left, marked with dots, can be thought of as a Ferrers dot diagram of a partition of an ordinary integer, in this case,  $6+5+4+3+2+1=21$ .



In finding the number of unsatisfactory paths, therefore, we are concerned with calculating the number of partitions which produce lines of route crossing the line  $y = x$ . For sake of convenience, let us call any partition (including 0) which would violate the ballot problem a  $k$ -partition. The zero partition must be included because an unsatisfactory path is produced

even if there are no dots at all. Using the ballot problem solution given above, the number of  $k$ -partitions for a rectangle of dimensions  $p$  and  $q$  is

$$\frac{qx}{p+1}$$

where  $x$  is the number of ordinary partitions (including 0) which can be drawn in a  $p \times q$  rectangle. The number of  $k$ -partitions for a rectangle of dimensions  $p+1$  and  $q$  is

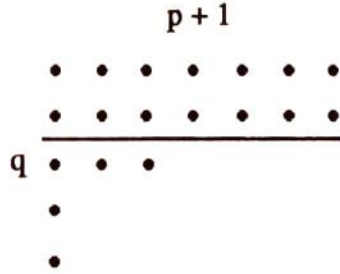
$$\frac{qx'}{p+2}$$

where  $x'$  is the number of partitions which can be drawn in a  $(p+1) \times q$  rectangle. This leads to the equation

$$\frac{qx}{p+1} + y = \frac{qx'}{p+2} \quad (1)$$

where  $y$  is the number of  $k$ -partitions with at least one part of magnitude  $p+1$ . We first wish to calculate the number of these additional partitions.

Let  $K(a,b)$  be the number of  $k$ -partitions in a lattice of  $a$  horizontal units and  $b$  vertical units. In this case,  $a=p$  and  $b=q$ . Any partition counted as part of  $y$  has at least one row with  $p+1$  dots. Since a partition with rows of  $p+1$  dots is not a  $k$ -partition unless it is a  $k$ -partition without those rows, we can neglect those rows as shown below:



The number of rows of  $p+1$  dots divides  $y$  into different groups. For example, if only one such row exists, the number of  $k$ -partitions that are possible if that row is removed is  $K(p,q-1)$ . We are thus led to the formula:

$$y = K(p,q-1) + K(p,q-2) + \dots + K(p,1).$$

However, by definition,

$$K(a,b) = \frac{b}{a+1} X$$

where  $X$  is the number of partitions, including the zero partition, which can be drawn in an  $a \times b$  rectangle. This leads to the equation

$$y = \frac{q-1}{p+1} x_1 + \frac{q-2}{p+1} x_2 + \dots + \frac{1}{p+1} x_{q-1}$$

where  $x_1, x_2, \dots, x_{q-1}$  represent the number of partitions (including 0) that can be drawn in rectangles of length  $p$  and width  $q-1, q-2, \dots, 1$  respectively. Recall that, by Equation 1,

$$y = \frac{qx'}{p+2} - \frac{qx}{p+1}.$$

We can conclude that:

$$\frac{q-1}{p+1} x_1 + \frac{q-2}{p+1} x_2 + \dots + \frac{1}{p+1} x_{q-1} = \frac{q}{p+2} x' - \frac{q}{p+1} x \quad (2)$$

Let us now consider the more general case of the total number of partitions which can be drawn in a rectangular lattice. In going from  $x$  to  $x'$ , we must add on any partitions which have at least one row of  $p+1$  dots. As before, we can consider separate cases for the number of such rows ranging from 1 to  $q$ . For example, if one such row exists, the number of additional partitions is the number of partitions which can be drawn in a lattice of dimensions  $p$  and  $q-1$ . We have already let  $x_1, x_2, \dots, x_{q-1}$  represent the number of partitions that can be drawn in rectangles of length  $p$  and width  $q-1, q-2, \dots, 1$  respectively. Also, there is a partition which fills up the entire  $p+1$  by  $q$  grid. Consequently:

$$x' - x = x_1 + x_2 + x_3 + \dots + x_{q-1} + 1. \quad (3)$$

Solving for  $x$ ,

$$x = x' - (x_1 + x_2 + x_3 + \dots + x_{q-1} + 1)$$

Upon substituting this into Equation (2), we arrive at

$$\frac{q-1}{p+1} x_1 + \frac{q-2}{p+1} x_2 + \dots + \frac{1}{p+1} x_{q-1} = \frac{q}{p+2} x' - \frac{q}{p+1} x' + \frac{q}{p+1} (x_1 + x_2 + x_3 + \dots + x_{q-1} + 1)$$

$$\frac{-1}{p+1} x_1 + \frac{-2}{p+1} x_2 + \dots + \frac{1-q}{p+1} x_{q-1} = \left( \frac{q}{p+2} - \frac{q}{p+1} \right) x' + \frac{q}{p+1}$$

$$\frac{-x_1 - 2x_2 - 3x_3 - \dots - (1-q)x_{q-1}}{p+1} = -\frac{q}{(p+1)(p+2)} x' + \frac{q}{p+1}$$

$$x_1 + 2x_2 + 3x_3 + \dots + (q-1)x_{q-1} = \frac{q}{p+2} x' - q \quad (4)$$

Let  $z_1$  be the number of partitions which can be graphed in a lattice of dimensions  $q-1$  and  $p+1$ . By substituting  $z_1$  for  $x'$  and  $x_1$  for  $x$ , it follows from (3) that

$$\begin{aligned} z_1 - x_1 &= x_2 + x_3 + \dots + x_{q-1} + 1 \\ z_1 &= x_1 + x_2 + x_3 + \dots + x_{q-1} + 1 \end{aligned} \quad (5)$$

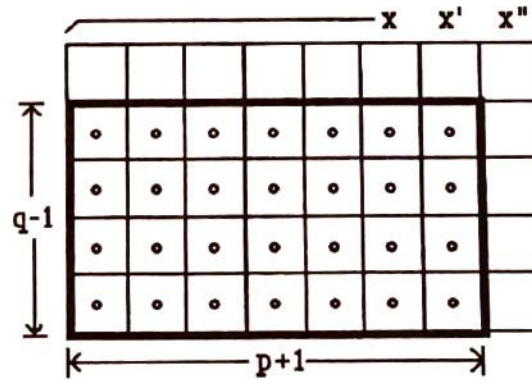
If  $z_2$  represents the number of partitions in a lattice of length  $p+1$  and height  $q-2$ , it is clear by similar reasoning that

$$z_2 = x_2 + x_3 + x_4 + \dots + x_{q-1} + 1.$$

Defining  $z_3, z_4$ , etc. in a similar fashion, taking their sum from  $q-1$  down to 1, adding 1 as in (3), and substituting (4) produces

$$\begin{aligned} z_1 + z_2 + z_3 + \dots + z_{q-1} + 1 &= x_1 + 2x_2 + 3x_3 + \dots + (q-1)x_{q-1} + q \\ z_1 + z_2 + z_3 + \dots + z_{q-1} + 1 &= \frac{q}{p+2} x' \end{aligned} \quad (6)$$

In the diagram below, all of the  $z$ -partitions are contained within the outlined box.



However, the left side of (6) is analogous to the right side of (3), except that instead of  $x' - x$  we have  $x'' - x'$ , where  $x''$  is the number of partitions in a grid of dimensions  $q$  and  $p+2$ . (See illustration.) Our final result is

$$\begin{aligned} x'' - x' &= \frac{q}{p+2} x' \\ x'' &= \frac{p+q+2}{p+2} x'. \end{aligned}$$

This conclusion can be stated more formally as follows.

**THEOREM.** If a lattice has dimensions of  $a$  horizontal and  $b$  vertical units ( $a > b$ ), and if  $N_{ab}$  represents the number of partition graphs that can be drawn in this lattice, including the 0-partition, then

$$N_{a+1,b} = \frac{a+b+1}{a+1} N_{ab}$$

where  $N_{a+1,b}$  is the number of graphs which can be drawn in the lattice when one vertical column is added.

We thus arrive at a recursive formula for the number of partitions which can be inscribed in a rectangular lattice. The formula also seems to work when  $a \leq b$ , but this would have to be proved separately, since we cannot simply interchange  $a$  and  $b$ .

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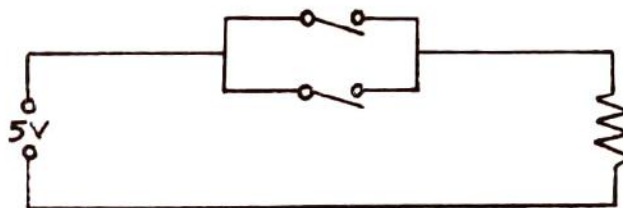
# A. P. SHOP

By Beata Geyer

It was September 11, 1989 - or was it? I felt as if, by walking through the door numbered 309, I went back to an earlier day when Stuyvesant was an all boys school. But wait there was Won-Hee, my ally; my only ally in a sea of twenty-two males until one more female transferred in. Truthfully, why did I pick digital electronics shop? It was a very lucky process of elimination. I wanted to keep my fingers intact. So, for all of you underclassmen who think you are leery of shop prospects and who are semi-logical, this is probably the best shop class the school has to offer.

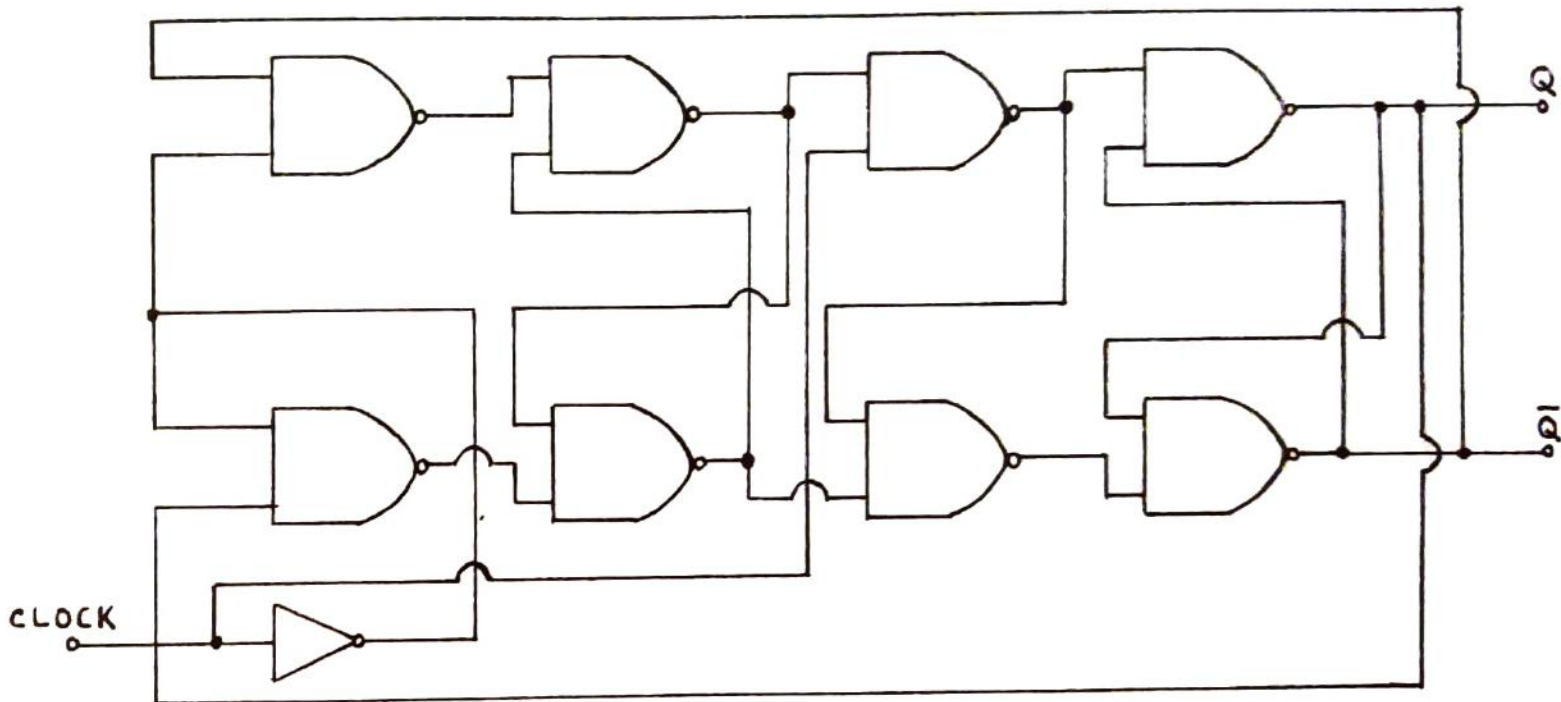
I had no idea of what to expect that fateful September day. I had even heard a rumor that we would be taking computers apart. We didn't quite get to that, but now I'm proud to say that I know why it's not just a magical experience when you turn it on. The most complex computer can eventually be explained through logic.

That's how we began. Remember, way back in Sequential Math I, when we all had to memorize the logic truth tables, when yes and no meant no, but yes or no meant yes, and no or no meant no, and not no and yes meant yes? These are the basics. By putting together a few switches, we are able to produce the desired result. For example, if you want to put together a circuit that performs the OR truth table, it's just a matter of two switches.



All logic gates are built with switches similar to these. Putting these gates together allows for projects of increasing complexity. Our first project was the full adder, which would add three inputs and leave us with the sum. It was one of the most difficult projects making us venture into the "black box" with such naivete. It was frustrating those nine tries when, despite having pulled all of our hair out, there would be some wrong connection. But

that tenth time was joyous. We had actually taken this mess of wires and put order into them. Everything is a puzzle or a flow-chart. It is easy to follow what this circuitry does as long as one knows the symbols. In the following example, the circuit does binary division because the clock has two states, high and low, or on and off. The clock is an input which changes rhythmically:



D NAND-Inverted and gates

▷ Inverter

Circuits like this one have thousands of applications. Computers, of course, are the most dominant. Electronics perform the operations and store information. So many household devices have digital displays and counters. Other applications include data transmission, synchronizers, electronic keyboards, alarms, and humidity gauges. The fields are endless. Digital electronics certainly opened my eyes. I never knew a shop could be so useful. So, underclassmen, don't let its name intimidate you. The thrill of victory when you do something technically correct is absolutely wonderful.

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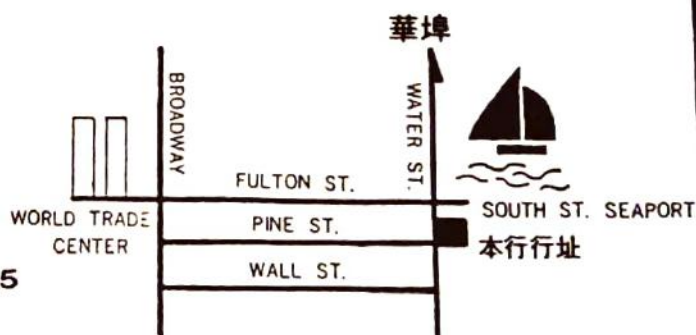
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# The Easter Egg Mystery

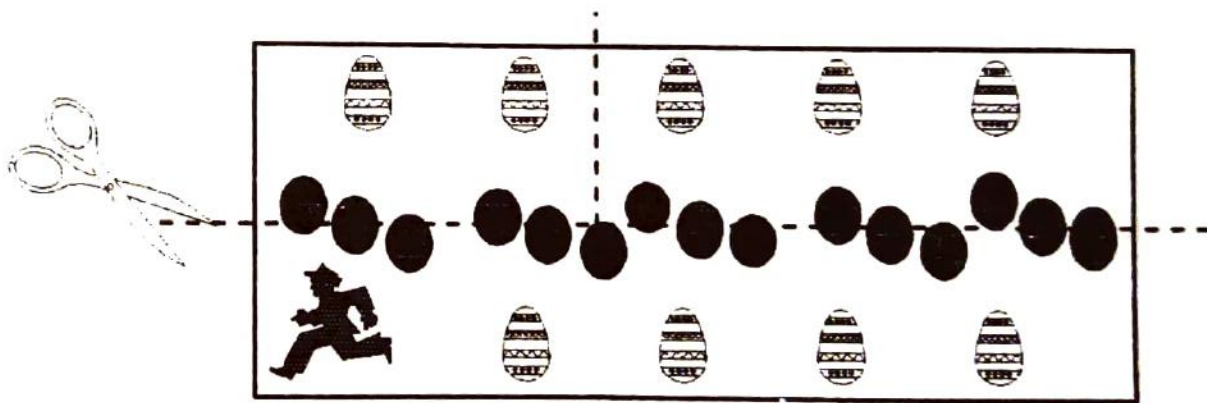
by David Nowakowski

## Directions:

1. Look carefully at the box, counting all the eggs.
2. Cut out the box, and then cut along the dotted lines.
3. Exchange the top two pieces.



4. Count the eggs. What happened?!?



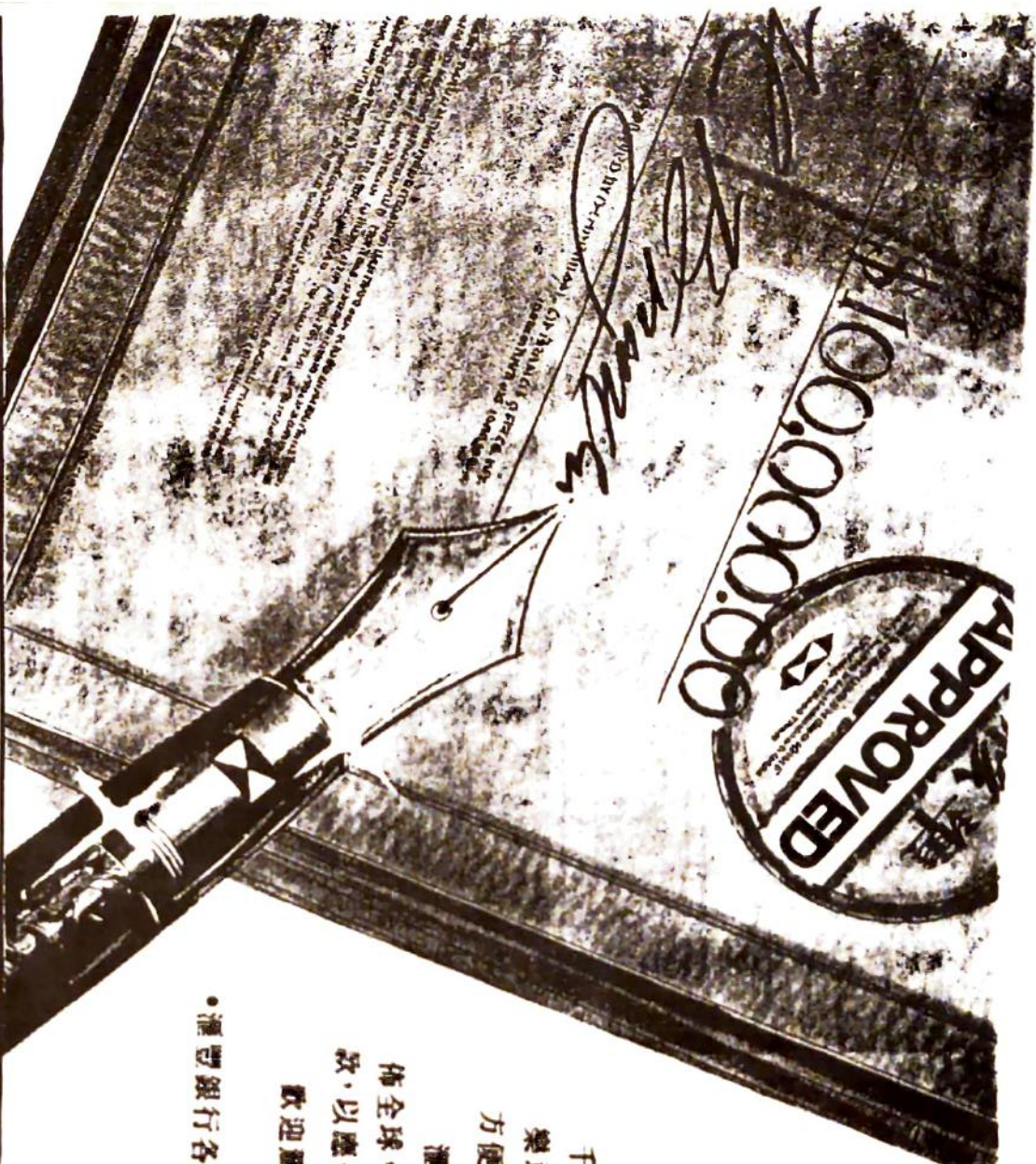
## Hints:

- a. Try to reexchange the pieces quickly. Try to see the missing egg returning.
- b. Repeat step three, keeping track of each egg.
- c. This is a math magazine! Try numbering the eggs.
- d. Ignore the first three hints - as you can see, they are no help at all.

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# Regents Refresher

Due to the New York State Board of Regents' introduction of a new high school mathematics curriculum, it is more difficult for us to provide overview of the material covered than in previous years. Readers who are in the classic courses—algebra, geometry, and trigonometry, designated M1/2, M3/4, and M51/52—should have no difficulty in recognising the material required for their exams. For the benefit those in sequential math—MQ1/2, MQ3/4, and MQ5/6—we have prefixed the appropriate sections with roman numerals, thus indicating the year in which the material is taught.

This section does not pretend to be a complete review. It includes only the most important theorems and formulæ.

## (II,III) Equations of Plane Curves:

Line:  $y=mx+b$  where  $m$ = slope and  $b$ = y-intercept.

$y-y_1=m(x-x_1)$  where  $(x_1,y_1)$  is a point on the line and  $m$  is the slope.

Slopes of parallel lines are equal.

The product of the slopes of two perpendicular lines is  $-1$ .

Circle:  $(x-h)^2 + (y-k)^2 = r^2$ , where  $(h,k)$  is the center of the circle with radius  $r$ .

Ellipse:  $x^2/a^2 + y^2/b^2 = 1$ , where  $a$  and  $b$  are the major and minor axes.

Parabola:  $(x-h)^2 = 2p(y-k)$ , where  $(h,k)$  is the vertex and  $p$  is the distance from the vertex to the directrix

or  $y = ax^2 + bx + c$ , where  $x = -b/2a$  is the axis of symmetry.

Hyperbola:  $x^2/a^2 - y^2/b^2 = 1$ , where  $a$  and  $b$  are the major and minor axes.

## (III) Logarithms: " $\log_b N=E$ " means $b^E=N$ .

$$\log_c 1=0 \quad \log_c c=1 \quad \log_c c^b=b$$

$$\log_c a^b=b \log_c a \quad \log_c ab=\log_c a + \log_c b \quad \log_c 1/a=-\log_c a$$

$$\log_c (a/b)=\log_c a-\log_c b$$

$$\log_c b^{\sqrt{a}}=(\log_c a)/b$$

$$\log_b a=\log_c a/\log_c b=\log_b c/\log_a c$$

## (II) Laws of Exponents:

$$x^m \cdot x^n = x^{m+n} \quad (x^m)^n = x^{mn} \quad x^m/x^n = x^{m-n} \quad x^{-m} = 1/x^m \quad x^0 = 1$$

$$x^{1/m} = m\sqrt{x} \quad x^{m/n} = n\sqrt{x^m} = (n\sqrt{x})^m$$

## (II) The Quadratic Formula:

The roots of the equation  $ax^2 + bx + c$  are

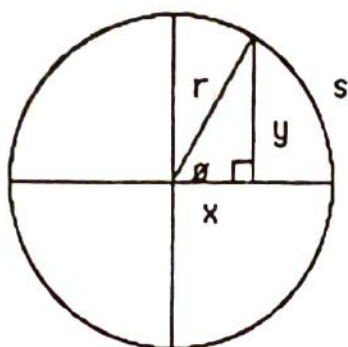
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

If the discriminant,  $b^2 - 4ac$ , is positive: the roots are real and distinct.

zero: the roots are real and equal.

negative: the roots are complex conjugates.

### (III) Trigonometry:



SOH-CAH-TOA

$$\sin \theta = y/r$$

$$\cos \theta = x/r$$

$$\tan \theta = y/x$$

$$\csc \theta = r/y = 1/\sin \theta$$

$$\sec \theta = r/x = 1/\cos \theta$$

$$\cot \theta = x/y = 1/\tan \theta$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$\cot^2 \theta + 1 = \csc^2 \theta$$

$$\sin \theta = \cos(90^\circ - \theta)$$

$$\tan \theta = \cot(90^\circ - \theta)$$

$$\sec \theta = \csc(90^\circ - \theta)$$

$$s = r\theta$$

$$r^2 = x^2 + y^2$$

Sum/Difference Formulæ:

$$\sin(x \pm y) = \sin x \cos y \pm \sin y \cos x$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

Double Angle Formulæ:

$$\sin(2x) = 2 \sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$= 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\tan(2x) = \frac{2 \tan x}{1 - \tan^2 x}$$

(Note: the double angle formulæ can be derived from the sum formulæ by letting  $x=y$ .)

Half Angle Formulæ:

$$\sin^2(x/2) = \frac{1 - \cos x}{2} \quad \cos^2(x/2) = \frac{1 + \cos x}{2}$$

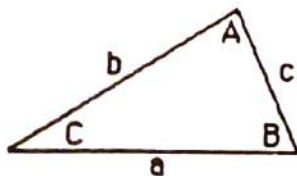
$$\tan^2(x/2) = \frac{1 - \cos x}{1 + \cos x}$$

In the graph of  $y = N \sin(rx)$ , the amplitude is  $N$ , and the period is  $2\pi/r$ .  
360 degrees =  $2\pi$  radians.

To convert from degrees to radians, multiply by  $\pi/180$ .

To convert from radians to degrees, multiply by  $180/\pi$ .

### Triangles:



$$\text{Area} = (1/2)ab \sin C$$

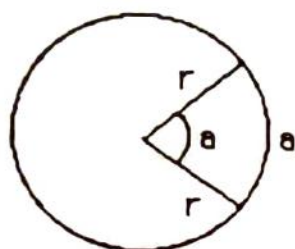
$$= (1/4)\sqrt{(a+b+c)(a+b-c)(b+c-a)(c+a-b)}$$

Law of Sines:  $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \text{diameter of circumcircle}$

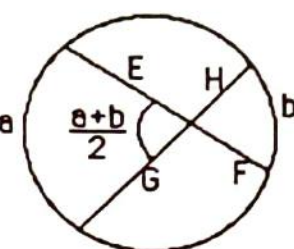
Law of Cosines:  $c^2 = a^2 + b^2 - 2ab \cos C$

Law of Tangents:  $\frac{\tan((A+B)/2)}{\tan((A-B)/2)} = \frac{a+b}{a-b}$

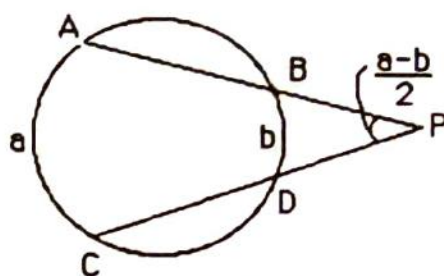
(III) Geometry:



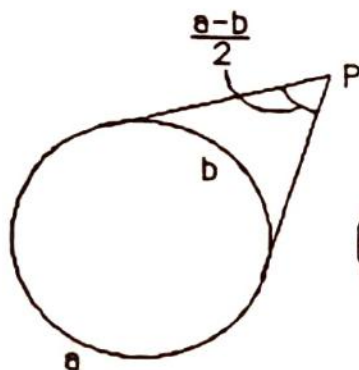
$r = \text{radius}$



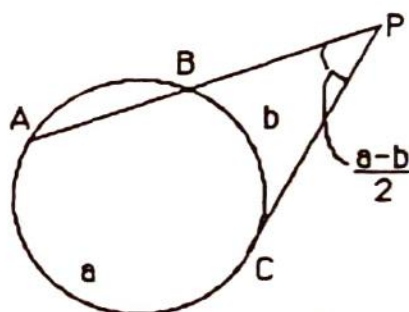
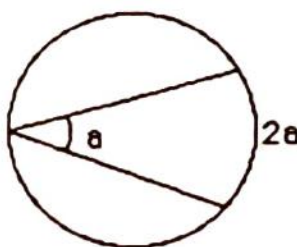
$EF = GH$



$(PA)(PB) = (PC)(PD)$



$PA = PB$



$(PA)(PB) = (PC)^2$

The required proofs for the Geometry Regents may be found in the Dressler text, pp. 733-737. The following are the numbers of the required proofs; 3,5,6,7,8,9,10,11,12,13.

(II) (A=Angle, S=Side, b=base, h=altitude)

Two triangles are congruent if you can prove SSS, SAA, SAS, or ASA.

Two triangles are similar if you can prove AA.

Two right triangles are congruent if you can prove their hypotenuses are congruent and one pair of legs are congruent. (HL)

The sum of the angles in a triangle is  $180^\circ$ .

The base angles of an isosceles triangle are congruent.

Area of triangle  $= (\frac{1}{2})bh$ .

Area of a trapezoid  $= (\frac{1}{2})(b_1 + b_2)h$ .

Area of a parallelogram  $= bh$ .

Area of a regular n-gon  $= (\frac{1}{2})ap$  (where  $a$ =apothem,  $p$ =perimeter).

In a right triangle with legs  $a$  and  $b$  and hypotenuse  $c$ ,  $a^2 + b^2 = c^2$ .

In a circle, the diameter that is perpendicular to a chord bisects the chord.

The locus of points at distance  $r$  from point  $P$  is the circle with center  $P$  and radius  $r$ .

The locus of points equidistant from two points  $P$  and  $Q$  is the perpendicular bisector of  $PQ$ .

The locus of points equidistant from two lines  $l$  and  $m$  are the bisectors of the angles formed by  $l$  and  $m$ .

( $B$ =area of base,  $h$ =height,  $r$ =radius)

Volume of a prism  $= Bh$

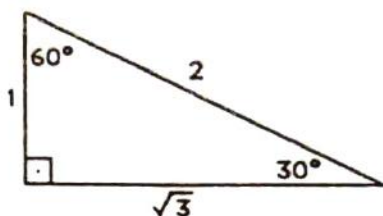
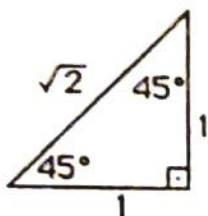
Volume of a cylinder  $= \pi r^2 h$

Volume of a pyramid  $= (\frac{1}{3}) Bh$

Volume of a cone  $= (\frac{1}{3}) \pi r^2 h$

Volume of a sphere  $= (\frac{4}{3}) \pi r^3$

The required constructions for the Geometry Regents may be found in the Dressler text, pp. 616-639. The following are the numbers of the required constructions; 5-7, 12, 13, 18-22, 25, 27-29.



(I, III) Probability:

Let A and B represent events in a sample space.

If A and B are independent, then  $P(A \cap B) = P(A) \cdot P(B)$ .

If A and B are disjoint, then  $P(A \cap B) = 0$  and  $P(A \cup B) = P(A) + P(B)$ .

If the probabilities are uniformly distributed, event A can occur in  $n(A)$  ways, and there are  $n(S)$  possible outcomes in S, then  $P(A) = n(A)/n(S)$ .

The number of permutations of n objects is  $n! = n(n-1) \cdots (2)(1)$ .

The number of permutations of n objects, k of which are indistinguishable, is  $n!/k!$ .

The number of permutations of r objects chosen from a set of n objects is

$${}_nP_r = n!/(n-r)! = n(n-1)(n-2) \cdots (n-r+1).$$

The number of combinations of r objects chosen from a set of n objects is

$${}_nC_r = n!/r!(n-r)! = {}_nP_r/r!.$$

(III) The binomial theorem:

$$(a+b)^n = \sum_{r=0}^n {}_nC_r a^r b^{(n-r)} = a^n + {}_nC_1 a^{n-1} b + {}_nC_2 a^{n-2} b^2 + \cdots + {}_nC_{n-1} a^1 b^{n-1} + b^n$$

(I,II) Logic:

Laws of Inference

Symbolic Forms

1. Law of Detachment  
(Modus Ponens)

$$[(p \rightarrow q) \wedge p] \rightarrow q$$

2. Law of the Contrapositive

$$(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$$

3. Law of Modus Tollens

$$[(p \rightarrow q) \wedge \sim q] \rightarrow \sim p$$

4. Chain Rule (Law of Syllogism)

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$$

5. Law of Disjunctive Inference

$$[(p \vee q) \wedge \sim p] \rightarrow q$$

$$[(p \vee q) \wedge \sim q] \rightarrow p$$

6. Law of Double Negation

$$\sim(\sim p) \leftrightarrow p$$

7. De Morgan's Laws

$$\sim(p \wedge q) \leftrightarrow (\sim p \vee \sim q)$$

$$\sim(p \vee q) \leftrightarrow (\sim p \wedge \sim q)$$

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# CLASSIFYING HAMILTONIAN LINES IN GRAPH THEORY

by Edmund Park

Graph theory, a branch of topology, has recently been through a great amount of development, as it can be applied to many fields including chemistry, physics, electronics, and of course, mathematics. A well known problem from graph theory is the traveling salesman problem, in which a salesman must visit each of  $n$  houses in a certain neighborhood, while traveling the shortest possible distance. Such a route was known to be called a Hamiltonian circuit, and was first proposed in a game developed by Sir William Rowan Hamilton.

Like the traveling salesman problem, the object of Hamilton's game was to travel through each of the vertices of a dodecahedron once, in a circuit that starts and ends at the same vertex. The important development from this problem was not the game itself, but the concept of a Hamiltonian line, or circuit. A Hamiltonian line in a graph is a circuit that passes through each of the vertices exactly once, and ends up at the origin point. It does not, in general, cover all the edges; in fact, it only covers two edges at each vertex.

One of the unsolved problems in graph theory is the classification of Hamiltonian circuits. However, great advancement was made in 1968, when a necessary condition for a planar graph to have a Hamiltonian circuit was presented at a conference on graph theory at Manebach, Germany. (Graphs which can be represented in a plane in such a way that no edges meet at a point other than a vertex are called planar graphs.) This was presented by the two Soviet mathematicians, Kozyrev and Grinberg.

If we call  $f(n)$  the number of  $n$ -gons in the interior of some questionable circuit  $C$ , and  $f'(n)$  the number of  $n$ -gons in the circuit's exterior, then, by the Kozyrev-Grinberg condition:

$$\sum_{n=3}^k (n-2)[f(n)-f'(n)] = 0.$$

The values of  $f(n)$  and  $f'(n)$  depend on the course taken through the graph by the circuit  $C$ . If  $C$  is the unknown object of one's investigations, the value of this condition would appear to be lost in the circularity of the situation. However, the equation involves  $n$  as well as  $f(n)$  and  $f'(n)$ , and this opening is all we need to utilize the theorem usefully.

The Kozyrev-Grinberg theorem, although somewhat simple, is a very important tool in characterizing Hamiltonian graphs. By simply using the formula given above, any planar graph may be tested for the existence of Hamiltonian circuits.

Let us now look at an example illustrating the Kozyrev-Grinberg theorem of Hamiltonian circuits. The given graph  $Q$  (Figure 1.) does possess various Hamiltonian circuits. How can we show, not by trial and error, that the graph has Hamiltonian circuits?

By the Kozyrev-Grinberg theorem,

$$2[f(4) - f'(4)] + 3[f(5) - f'(5)] = 0.$$

This implies that  $f(4) - f'(4)$  must be divisible by 3. Looking at the graph we see that that there are five 4-gons. Thus if  $f(4) = 4$  and  $f'(4) = 1$  then we have a case that works. (see Figure 1a.)

$Q$ :

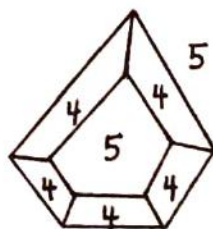


Figure 1

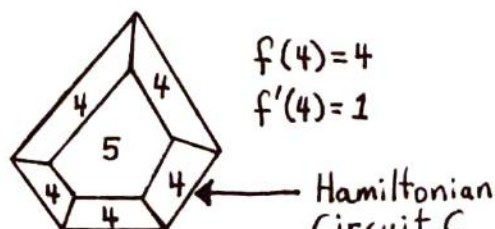


Figure 1a

# LINEAR DIOPHANTINE EQUATIONS

by Albert Pahk

**Problem:** In a corral, there are a number of cowboys and an odd number of horses. It is observed that there are 20 legs in all. How many legs belong to the horses?

This is a typical linear diophantine equation. Linear diophantine equations are equations with two or more variables whose solutions are positive integers. It was first considered centuries ago in Alexandria by an Egyptian mathematician named Diophantus. The form for a general linear diophantine equation is,

$$ax + by = c, \text{ where } a, b, c \text{ are all integers.}$$

For a mathematician ignorant of modular arithmetic, the sensible approach would be substitution.

$$4h = 20 - 2c \text{ or } 2h = 10 - c$$

Using the method of trial and error by substituting for  $c$ , we get a infinite number of solutions with no method to ascertain a pattern. However, by using congruences, we can find a pattern. The best method for solving linear diophantine equations is the method of linear congruences.

Congruences can be either linear, quadratic, cubic, or any other power. The general form for a linear congruence is  $ax \equiv b \pmod{m}$ . When dealing with these equations, the problem solver wants to know immediately whether or not it has a solution and if so how many solutions it has. There are lemmas that help determine such information.

**Lemma I:** If  $(a, m) \nmid b$ , then  $ax \equiv b \pmod{m}$  has no solutions. This is read: if the greatest common divisor of  $a$  and  $m$  does not divide  $b$ , then  $ax \equiv b \pmod{m}$  has no solutions.

**Lemma II:** If  $(a, m) = 1$ , then  $ax \equiv b \pmod{m}$  has exactly one solution. This is read: if the greatest common divisor of  $a$  and  $m$  is 1, then  $ax \equiv b \pmod{m}$  has exactly one solution.

**Lemma III:** Letting  $d = (a, m)$ , if  $d \mid b$ , then  $ax \equiv b \pmod{m}$  has exactly  $d$  solutions. This is read: if the greatest common divisor of  $a$  and  $m$  equals  $d$  and if  $d$  divides  $b$  then there are exactly  $d$  solutions.

Once the number of solutions is determined, trial and error must be applied in finding the actual solutions. Unless the modulus and the residue are very large, finding the solutions should not be hard because the answers will be in the form of an arithmetic progression.

Solution: The equation  $ac + bh = z$ , can be split into two different congruences of either  $ac \equiv z \pmod{b}$  or  $bh \equiv z \pmod{a}$ . In this problem, where  $a=2$ ,  $b=4$ ,  $z=20$  and  $c$  and  $h$  are the unknowns, the solution could be most easily found in the form  $2c \equiv 20 \pmod{4}$ . the equation simplifies into  $c \equiv 0 \pmod{2}$ , which implies that the solution for  $c$  is the set of all even integers. Normally, in order to find the different values of  $c$  a table would be necessary. Once a solution for  $c$  is found, it must be substituted back into the original equation to see if  $h$  has an integral solution as well. Even though this requires some trial and error, it has an easier pattern to follow than random guessing.

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# ERROR-CORRECTING CODES

by Nina Yuan

Error-correcting codes are algorithms developed to detect or correct errors that may occur in the transmission of data. They are used not only for communications channels, such as radio communications via satellite, but also to correct errors in data stored on a magnetic medium, such as a computer diskette.

The "goodness" of a code is determined by its rate, which is defined as the ratio of information bits to total bits required to send a given message. If code  $A$  has a larger ratio than code  $B$ , then code  $A$  has the ability to detect or correct more error, and therefore theoretically, it is a "better" code than  $B$ . In practice, however, considerations such as the ease of encoding and decoding must also be taken into account.

For a code  $C$ , the weight,  $wt(x)$ , of a code word  $x$  is defined as the number of nonzero components of  $x$ . The distance between two code words

$$x, y \in C$$

equals the number of components which differ, that is,

$$D(x, y) = |\{i: x_i \neq y_i\}|$$

The minimum nonzero weight of the code can be proven to equal the minimum distance,  $d$ , between any two distinct code words. It can also be proven that a code  $C$  with a minimum nonzero weight,  $d$ , can correct up to  $d-1$  transmission errors and correct a maximum of  $(d-1)/2$  errors. Obviously, codes with larger values of  $d$  have code words that are more distinct and thus can correct and detect a greater number of errors.

The simplest case to consider when designing a code is a binary symmetric channel (B.S.C) over which only 0's or 1's are transmitted. Assume that a transmitted digit will be correctly received with a probability  $q$ ; therefore, the probability that it will be incorrectly received is  $1-q$ . The search for new codes is motivated by Shannon's Theorem, which states that, for a given time limit,  $T$ , a code for transmitting  $n$  information bits can be constructed

such that it has an arbitrarily small error probability.

One obvious method of insuring the accurate transmission of data is to transmit it more than once. However, such a redundancy code is very inefficient in terms of time and medium storage space. If each bit  $a$  is sent  $n$  times, then the receiver can decode correctly only if there are less than  $n/2$  errors. The inefficiency arises from the fact that  $1/n$  of each codeword is information, and  $n-1$  bits are redundant. A better method to detect errors is to perform a parity check. If  $n$  information bits  $a_1, a_2, \dots, a_n$  are to be transmitted, then

$$a_{n+1} = a_1 + a_2 + \dots + a_n \pmod{2}.$$

If the sum of the received bits

$$b_1 + b_2 + \dots + b_n = b_{n+1}$$

then the receiver can conclude that there were zero or an even number of errors. Otherwise, there were an odd number of errors. Obviously, this code works best when the probability of more than one error occurring is very small. It is very efficient, since only one redundancy bit is transmitted, and  $n/(n+1)$  of the code is information.

The parity check is widely used in computer memory chips to detect errors. However, it can only detect single errors and cannot correct multiple errors. A more useful code is the Hamming (7,4) code, where 4 information bits and 3 redundancy bits are transmitted. We begin by defining

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

which has the property that all columns of  $H$  are different. To encode an information word

$(a,b,c,d)$ , let

$$C = \begin{bmatrix} x \\ y \\ a \\ z \\ b \\ c \\ d \end{bmatrix}$$

so that the matrix multiplication of  $H$  and  $C$  equals 0,  $HC = 0$ ; in other words,

$$x + a + b + d = 0$$

$$y + a + c + d = 0$$

$$z + b + c + d = 0.$$

If

$$R = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \\ b_6 \\ b_7 \end{bmatrix}$$

is received and  $R = C$ , by computing  $HR = 0$ , the receiver knows that  $HC = 0$  and the message was received correctly. If one error occurs, then  $HR = HC + HE = 0 + HE$ , where  $E = R - C$ . Once  $E$  is known,  $C$ , and thus the original message, can be determined. If more than one error occurs, the fact that an error has occurred can be detected, but the errors cannot be corrected because  $A$  and  $B$ , where  $E = A + B$ , cannot be uniquely

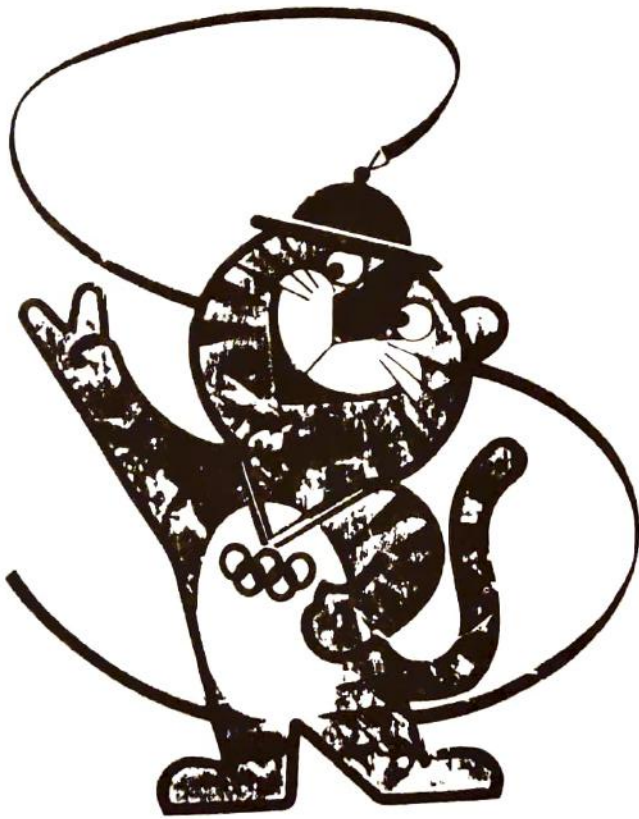
determined.

This simple code can be quite effective. For example, if a message is transmitted without an error-correcting code on a binary channel where the probability of receiving a bit correctly is  $q=0.90$ , then the probability of receiving a four-bit message is  $q^4=0.6561$ . By using the (7,4) Hamming code, the probability of receiving a four-bit message correctly is the sum of the probability of no errors and the probability of only one error, or  $q^7 + 7pq^6 = 0.4783 + 0.3720 = 0.8503$ . For real-world applications, however, this code is much too limited. Although improved Hamming codes have been developed, more progress has been made with codes based on finite field theory, such as quadratic residue codes and B.C.H. codes (developed by Bose, Chaudhuri, and Hocquenghem), which can correct multiple errors.

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# PELL EQUATIONS

By Yulia Shapiro

Pell equations were considered by many mathematicians and were first mentioned by Archimedes. The name for this special type of diophantine equation originated in the erroneous notion of Euler that John Pell gave the unique method for solving this type of equation. Although Pell contributed nothing to the subject, the name has remained.

The equation has the form:

$$x^2 - dy^2 = \pm 1.$$

The first published proof for solving the equation appeared in the 1657 by Lord Brouncker. He used continued fractions to solve the equation.

Continued fractions has the form:

$$\theta = A_0 + \frac{1}{A_1 + \frac{1}{A_2 + \frac{1}{A_3 + \dots}}}$$

Another way to express continued fractions is:  $\theta = [A_0, A_1, A_2, A_3, \dots]$ .  $\theta$  is always an irrational root. Therefore,  $\theta'$  is defined as its conjugate or the other root of the equation.  $p_n/q_n$  is defined as the convergent to  $\theta$  such that the continued fraction is calculated up to  $A_n$ . The following is the relationship for the subsequent  $p_n$  and  $q_n$ :

$$\frac{p_n}{q_n} = \frac{A_n p_{n-1} + p_{n-2}}{A_n q_{n-1} + q_{n-2}}$$

Also, the identity

$$p_n q_{n-1} - p_{n-1} q_n = \pm 1$$

is true for any  $p$  and  $q$ . Therefore the convergences follow the pattern:

$$p_0/q_0 < p_2/q_2 < p_4/q_4 < \dots < \theta < \dots < p_5/q_5 < p_3/q_3 < p_1/q_1.$$

Periodic continued fractions have the form:

$$\alpha = [A_0, A_1, \dots, A_{k-1}, \overline{A_k, \dots, A_{k+m-1}}]$$

where the bar indicated the repeating block. It can be proven that a continued fraction represents a quadratic irrational if and only if it is ultimately periodic. Since  $\sqrt{d}$  is a quadratic irrational it is obvious that it can be represented as a periodic continued fraction.

Any solution in positive integers  $x, y$  to the Pell equation satisfies the equation:

$$x - y\sqrt{d} = \frac{1}{x + y\sqrt{d}}.$$

Therefore,  $x > y\sqrt{d}$  and  $x - y\sqrt{d} < 1/2y\sqrt{d}$ . This gives  $|\sqrt{d} - x/y| < 2y^2$ , and it follows that  $x/y$  is convergent to  $\sqrt{d}$ . Therefore,  $x = p_n$  and  $y = q_n$  for some  $n$ , such as considered above. Considering the equation for  $+1$ , we note that  $n$  must be odd. In fact, it has to be in the form  $km-1$ , where  $k$  is a positive integer when  $m$  is even and where  $k$  is an even positive integer when  $m$  is odd. After working with different  $\theta$ 's and  $A$ 's, it follows that  $n+1$  is divisible by  $m$  and so  $n$  has the form  $km-1$  as asserted. This shows that the only positive solutions  $(x, y)$  to the Pell equations are given by  $x = p_n, y = q_n$  where  $p_n/q_n$  is convergent to  $\sqrt{d}$  with  $n$  of the form  $km-1$  as stated above.

All of the  $p_n$  and  $q_n$  satisfy the equation and thus comprise the full set of solutions. In view of periodicity of  $\sqrt{d}$ ,  $\theta_1 = \theta_{n+2}$  for all  $n = km-1$ . Hence,

$$\sqrt{d} = \frac{(p_{n+1}\theta_1 + p_n)}{(q_{n+1}\theta_1 + q_n)}$$

$$\sqrt{d} = \frac{A_0 + 1}{\theta_1}$$

$$p_n - q_{n+1} - A_0 q_n$$

$$p_{n+1} - A_0 p_n - q_n d$$

$$p_n^2 - dq_n^2 - p_n q_{n+1} - p_{n+1} q_n$$

Since  $n$  is odd, it follows that  $p^2 - dq^2 = 1$ .

With similar analysis for the equation  $x^2 - dy^2 = -1$ , we note that there are no solutions when  $m$  or the period of  $\sqrt{d}$  is even. When  $m$  is odd all positive solutions are given by  $x = p_n, y = q_n$  where  $p_n/q_n$  is convergent to  $\sqrt{d}$  when  $n = km - 1$  and  $k$  is an odd positive integer. When the equation is solvable, there is a solution of the smallest value,  $(x', y')$ , known as the fundamental solution, but this is not  $(\pm 1, 0)$  which is considered the trivial solution. With careful analysis of the equation  $\phi = x' + y'\sqrt{d}$ , one may deduce that all solutions are given by  $x + y\sqrt{d} = \pm \phi^n$ , where  $n$  is an odd integer. The result is easily obtained by noting that the fundamental solution to  $x^2 - dy^2 = 1$  is given by setting  $\epsilon = \phi^2$ . An analogous result is easily obtained to the more general equation  $x^2 - dy^2 = k$ , where  $k$  is a non-zero integer. If the equation is solvable, one can specify a finite set of solutions  $(x', y')$ , such that  $x + y\sqrt{d} = \pm \phi \epsilon^n$  where  $n$  is an even integer yields all solutions to  $\phi = x' + y'\sqrt{d}$ .

As an example, let us consider the equation  $x^2 - 97y^2 = -1$ . The continued fraction for  $\sqrt{97}$  is  $[9, 1, 5, 1, 1, 1, 1, 1, 5, 1, 13]$ . Thus the period of  $\sqrt{97}$  is 11 and since this is odd, the equation is solvable. The fundamental solution is given by  $x = p_{10}$  and  $y = q_{10}$ . The first convergents to 97 are  $10, 59/6, 69/7, 128/13, 197/20, 325/20, 522/53, 847/86, 4757/483, 5604/569$ . Hence the fundamental solution to  $x^2 - 97y^2 = -1$  is  $x = 5604$  and  $y = 569$ . Further if we let  $\phi = 5604 + 569\sqrt{97}$ , then  $\epsilon = \phi^2$  gives the fundamental solution to  $x^2 - 97y^2 = 1$ . The solution is in fact  $x = 62,809,633$  and  $y = 6,377,352$ .

Incidentally, the continued fraction for  $d$  always has the form:

$$[A_0, \overline{A_1, A_2, \dots, A_2, A_1, 2A_0}].$$

Moreover, the period of  $\sqrt{d}$  is always odd when  $d$  is a prime such that  $d \equiv 1 \pmod{4}$ . For all such  $d$ , the equation is always solvable.

The other method for solving the equation requires that the fundamental solution be known. Then the formula for finding every next solution is:

$$(x, y) \Rightarrow (ax + by, bx + ay),$$

where  $(a, b)$  is the fundamental solution. This formula may be derived after working with some general  $d$  and then by generalizing the results.

A problem for further consideration might be Pell equations to the powers other than two. On the first consideration those equations are not solvable, but a proof for the non-existence of solutions is not obvious.

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# THE NON-EXISTENCE OF ALMOST PERFECT NUMBERS OF CERTAIN FORMS

by Benjamin Bromberg

An almost perfect number is a number the sum of whose factors is twice the number minus unity (one). All powers of two are almost perfect and the existence of almost perfect numbers of a different form has been posed as a problem. The following proofs will demonstrate that almost perfect numbers not of the form  $2^n$ , if any exist, must either be or have as a factor an odd square with at least two distinct odd prime factors.

For all positive integers  $n$ ,  $\sigma(n)$  represents the sum of all positive divisors of  $n$ . A number is almost perfect if  $\sigma(n) = 2n - 1$ . All powers of two are almost perfect because the sum of its factors is  $2^{n+1} - 1$ , which is twice  $2^n$  minus unity. There have been numerous studies on the existence or non-existence of quasiperfect numbers,  $\sigma(n) = 2n + 1$ , but very little has been done on almost perfect numbers that are not powers of two. The following theorems may shed some light.

Note:  $p$  represents an odd prime throughout this article.

**Theorem I:** If  $k$  is almost perfect and of the form

$$2^{n_0} p_1^{n_1} p_2^{n_2} \dots p_s^{n_s}$$

where  $n_s$  is a natural number, then all  $n_s$ , for  $s \geq 1$ , must be even.

**Proof:** Since  $k$  is almost perfect;

$$\sigma(k) - 2k - 1$$

By definition:

$$\sigma(k) - (2^{n_0+1} - 1) \left( \frac{p_1^{n_1+1} - 1}{p_1 - 1} \right) \left( \frac{p_2^{n_2+1} - 1}{p_2 - 1} \right) \dots \left( \frac{p_s^{n_s+1} - 1}{p_s - 1} \right)$$

Therefore,

$$2k-1-(2^{n_0+1}-1)\left(\frac{p_1^{n_1+1}-1}{p_1-1}\right)\left(\frac{p_2^{n_2+1}-1}{p_2-1}\right)\dots\left(\frac{p_s^{n_s+1}-1}{p_s-1}\right)$$

By factoring, we can rewrite the over equation as;

$$2k-1-(2^{n_0}-1)(1+p_1+\dots+p_1^{n_1})\dots(1+p_1+\dots+p_s^{n_s})$$

The left side of this equation must be odd, therefore, the right side must be the product of odd numbers. Since the number of members (all odd) of each  $s^{\text{th}}$  term on the right is equal to  $n_s+1$ , if  $n_s$  is odd then the  $s^{\text{th}}$  term will be even. Since no term on the right can be even,  $n_s$  cannot be odd. Hence the theorem follows.

**Theorem II:** An odd almost perfect number must have at least two distinct odd prime factors.

**Proof:** Let  $k$  be an almost perfect number. Let  $k$  be odd and have one distinct odd prime factor. Hence  $k=p^n$ . By Theorem I,  $n$  must be divisible by two, so let  $n=2\alpha$ ;  $k=p^{2\alpha}$ .

Therefore,

$$2p^{2\alpha}-1-\sigma(p^{2\alpha})-\frac{p^{2\alpha}-1}{p-1}-(1+p+\dots+p^{2\alpha})$$

Hence, we can write:

$$2p^{2\alpha}-1-1+p+\dots+p^{2\alpha}$$

If we subtract  $p^{2\alpha}$  from both sides, then

$$p^{2\alpha}-1-1+p+\dots+p^{2\alpha-1}=(1+p)(1+p^2+\dots+(p^2)^{\alpha-1})$$

By dividing  $1+p$  from both sides, we get:

$$(p^{2\alpha-1}+p^{2\alpha-3}+\dots+p)-(1+p^2+\dots+(p^2)^{\alpha-1})-1+p^2+\dots+(p^2)^{\alpha-1}$$

We can rewrite our last equation as:

$$p^{2\alpha-1} + p^{2\alpha-3} + \dots + p - 2(1 + p^2 + \dots + (p^2)^{\alpha-1})$$

Because  $p$  is at least 3, each term on the left is greater than each respective term on the right;  $p^{2\alpha-m} > 2(p^2)^{\alpha-1}$ . Therefore, the above equality does not hold, and hence, we have proven theorem II by contradiction.

**Theorem III:** An even almost perfect number (not of the form  $2^n$ ) must have at least two distinct odd prime factors.

**Proof:** Let  $k$  be an almost perfect number. Let  $k$  be even and have one distinct odd prime factor. Hence  $k = 2^m p^n$ . By Theorem I,  $n$  must be divisible by two, so let  $n = 2\alpha$ ;  $k = 2^m p^{2\alpha}$ . Therefore,

$$2(2^m p^{2\alpha}) - 1 - \sigma(2^m p^{2\alpha}) - (2^{m+1} - 1) \left( \frac{p^{2\alpha} - 1}{p - 1} - (2^{m+1} - 1)(1 + p + \dots + p^{2\alpha}) \right)$$

By distributing, we get:

$$2(2^m)(p^{2\alpha}) - 1 - (2^{m+1})(1 + p + \dots + p^{2\alpha}) - (1 + p + \dots + p^{2\alpha})$$

By dividing by  $2^m$ , we get:

$$2(p^{2\alpha}) - \frac{1}{2^m} - 2(1 + p + \dots + p^{2\alpha}) - \frac{(1 + p + \dots + p^{2\alpha})}{2^m}$$

By subtracting  $2p^{2\alpha}$  from both sides, we get:

$$-\frac{1}{2^m} - 2(1 + p + \dots + p^{2\alpha-1}) - \frac{1 + p + \dots + p^{2\alpha}}{2^m}$$

We can further reduce this to:

$$\frac{(1 + p + \dots + p^{2\alpha}) - 1}{2^m} - \frac{p + \dots + p^{2\alpha}}{2^m} - 2(1 + p + \dots + p^{2\alpha-1})$$

The above equation can be further simplified to:

$$1+p+\dots+p^{2^a}-2^{m+1}(1+p+\dots+p^{2^a-1})$$

If we divide both sides by  $1+p+\dots+p^{2^a-1}$ , we get:

$$\frac{p+\dots+p^{2^a}}{1+p+\dots+p^{2^a-1}} = \frac{p(1+p+\dots+p^{2^a-1})}{1+p+\dots+p^{2^a-1}} - 2^{m+1}$$

This implies:

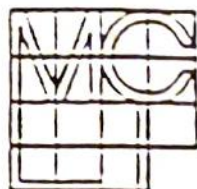
$$p-2^{m+1}$$

This is a contradiction since  $p$  is an odd prime. Hence, we have proven the theorem by contradiction. This proof can be extended to some degree to pertain to even numbers with more than one odd prime factor, but will not work completely because the product of the sums of the prime numbers may or may not be divisible by a common factor, as was the case here.

We see from the above proofs that almost perfect numbers that are not powers of two are limited, if not non-existent. Whether the number is even or odd it must have an odd square (with at least two prime factors) as a factor. Whether almost perfect numbers are always even or sometimes odd remains an area for further research.

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# THE TAUTOCHRONE

by Edward Valentini

A problem of interest in the history of mathematics is that of the tautochrone -- the curve down which a particle will slide freely under the force of gravity. It will reach the bottom at the same time regardless of its starting point on the curve.

This problem arose in the construction of a clock pendulum whose period is independent of the amplitude of its motion. The tautochrone was found by Christian Huygens (1629-1695) by geometrical methods.

If a particle is placed on the curve and is allowed to fall freely, the velocity of the particle is given by:

$$v = \frac{ds}{dt} \quad (i)$$

By the principle of conservation of energy:

$$\frac{1}{2}mv^2 - \frac{1}{2}mv_i^2 = mg(b - y) \quad (ii)$$

However, it is assumed that the initial velocity is zero and that the particle starts from rest. Eq.(ii) reduces to:

$$v = \sqrt{2g(b - y)} \quad (iii)$$

Combining (i) and (iii) we have:

$$\begin{aligned} \frac{ds}{dt} &= \sqrt{2g(b - y)} \\ dt &= \frac{ds}{\sqrt{2g(b - y)}} \end{aligned} \quad (iv)$$

$$ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

But

Therefore the time  $T(b)$  as a function of  $b$ , the ordinate of the starting point, is given by:

$$T(b) = \frac{1}{\sqrt{2g}} \int_0^b \sqrt{\frac{1 + \left(\frac{dx}{dy}\right)^2}{b-y}} dy \quad (v)$$

and letting:

$$f(y) = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}$$

$$T(b) = T_o \quad (vi)$$

Eq.(v) reduces to:

$$\sqrt{2g} T_o = \int_0^b \frac{f(y)}{\sqrt{b-y}} dy$$

Taking the Laplace transform of both sides:

$$\mathcal{L}\{\sqrt{2g} T_o\} = \mathcal{L}\left\{\int_0^b \frac{f(y) dy}{\sqrt{b-y}}\right\} \quad (vii)$$

results in:

$$\frac{\sqrt{2g} T_o}{s} = \frac{F(s) \Gamma\left(\frac{1}{2}\right)}{\sqrt{s}} \quad (viii)$$

where  $\Gamma(1/2)$  denotes the gamma function. The gamma function is defined by:

$$\Gamma(x) = \int_0^\infty e^{-t} t^{x-1} dt$$

Since  $\Gamma(1/2) = \pi^{1/2}$  we have, from Eq.(viii):

$$\frac{\sqrt{2g} T_o}{s} = F(s) \sqrt{\frac{\pi}{s}} \quad (ix)$$

---

<sup>1</sup>The proof is trivial and is therefore left for the reader.

or

$$F(s) = \sqrt{\frac{2g}{\pi}} \frac{T_o}{\sqrt{s}}$$

Letting

$$k = \sqrt{\frac{2g}{\pi}} T_o \quad (x)$$

Then:

$$F(s) = \frac{k}{\sqrt{s}} \quad (xi)$$

But the inverse Laplace:

$$\mathcal{L}^{-1} \left\{ k \sqrt{\frac{\pi}{s}} \right\} = \mathcal{L}^{-1} \frac{k}{\sqrt{\pi}} y^{-\frac{1}{2}} \quad (xii)$$

Therefore:

$$f(y) = \sqrt{\frac{2g}{\pi}} \frac{1}{\sqrt{\pi}} \frac{1}{\sqrt{y}} T_o$$

$$f(y) = \frac{\sqrt{2g}}{\pi} \frac{T_o}{\sqrt{y}} \quad (xiii)$$

From Eqs.(vi) we have:

$$\sqrt{1 + \left( \frac{dx}{dy} \right)^2} = \frac{\sqrt{2g} T_o}{\pi \sqrt{y}} \quad (xiv)$$

Upon rearranging terms and solving for dx/dy  
We are left with:

$$\frac{dx}{dy} = \sqrt{\frac{\left( \frac{2g T_o^2}{\pi^2} \right) - y}{s}} \quad (xv)$$

Replacing  $\frac{2gT_o^2}{\pi^2}$  by  $\alpha$  we have:

$$\frac{dx}{dy} = \sqrt{\frac{2\alpha - y}{y}} \quad (\text{xvi})$$

To solve this differential equation it is helpful to use the substitution:

$$y = 2\alpha \sin^2\left(\frac{\theta}{2}\right) \quad (\text{xvii})$$

Therefore:

$$\frac{dx}{dy} = \sqrt{\frac{2\alpha - 2\alpha \sin^2\left(\frac{\theta}{2}\right)}{2\alpha \sin^2\left(\frac{\theta}{2}\right)}} \quad (\text{xviii})$$

with the use of trigonometric identities Eq.(xviii) reduces to:

$$\frac{dx}{dy} = \cot\left(\frac{\theta}{2}\right) \quad (\text{xix})$$

which is simplified by inverting:

$$\frac{dy}{dx} = \tan\left(\frac{\theta}{2}\right) \quad (\text{xx})$$

From Eq.(xvii), we have

$$\begin{aligned} \frac{dy}{d\theta} &= \frac{4\alpha \sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}{2} = \alpha \sin \theta \\ dy &= \alpha \sin \theta d\theta \end{aligned} \quad (\text{xxi})$$

Since we have assumed that the particle started from rest we have the initial conditions:

$$y(0) = 0 \quad x(0) = 0 \quad (\text{xxii})$$

Integrating Eq.(xxi) and employing condition (xxii) we have:

$$y = a(1 - \cos \theta) \quad (\text{xxiii})$$

combining Eqs.(xx) and (xxi) we have:

$$\frac{\alpha \sin \theta d\theta}{dx} = \frac{1 - \cos \theta}{\sin \theta} \quad (\text{xxiv})$$

Simplifying we have:

$$\frac{dx}{d\theta} = \frac{\alpha \sin^2 \theta}{1 - \cos \theta} \quad (\text{xxv})$$

Upon further simplification we have:

$$dx = \alpha(1 + \cos \theta)d\theta \quad (\text{xxvi})$$

From Eqs.(xxvi) and (xxvii) and integrating we have:

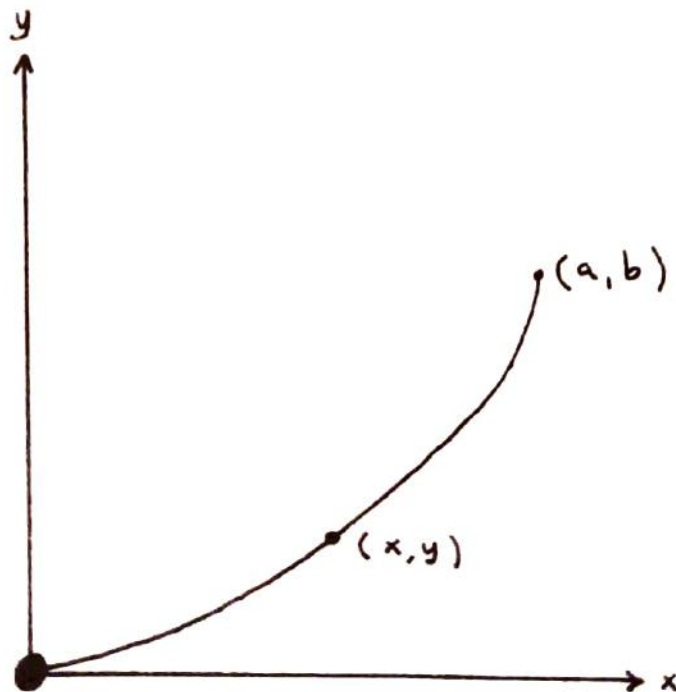
$$x = \alpha(\theta + \sin \theta) \quad (\text{xxvii})$$

From the previous assumptions, we have the final result. The parametric equations for the tautochrone are:

$$x = \frac{2 g T_o^2}{\pi^2}(\theta + \sin \theta)$$

$$y = \frac{2 g T_o^2}{\pi^2}(1 - \cos \theta)$$

which together define a cycloid called the tautochrone.



# Newton Learns About Calculus:

## A Math Team Classic

Told by Richard Sun

"No, no, no!," cried the Doctor in disgust. "You can't integrate the function that way." The Doctor was beginning to think that teaching Mr. Newton the Calculus was more trouble than escaping in the Tardis from the gravitational field of a densely packed F train.

"This Calculus is great! I wish I had discovered it myself," exclaimed Newton, who was bubbling over with joy. "Imagine," he pondered, "if I had made such a discovery. All of England would be amazed!"

"Oh, quite more than that, Isaac. I tell you what, though. You can take the credit for it. It's not very advanced mathematics from where I come from."

"But I couldn't..."

"Of course you can. No one will ever know."

"I can use this for *everything*! I could even find the volume of my teapot!"

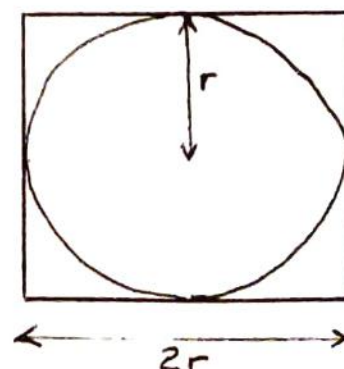
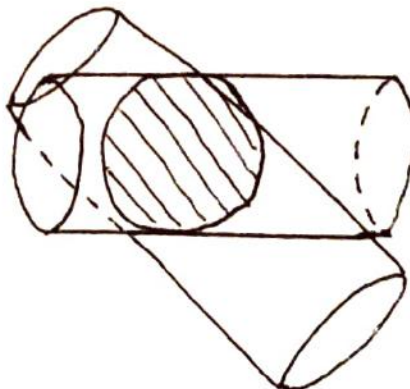
"Well I'm sure you'll do that later, Isaac. But you know, you just cannot use Calculus everywhere; you have to be discriminating at times."

"Really?," the puzzled Newton began, "I thought Calculus could solve any math problem."

"Well, there are just some problems where you wouldn't want to use Calculus. For example, you wouldn't use Calculus to find the area of a right triangle, now would you? I suppose not. A certain Greek acquaintance of mine, a man by the name of Archimedes — oh, you know him too? — well, he couldn't solve this silly problem involving the volume of intersection of two right circular cylinders of equal radius  $r$  whose axes of symmetry intersect at right angles to each other. I had to tell him how to do it:"

"First, we inscribe a sphere in the volume of intersection."

"Then, we look at cross sections of the combined figures which are taken parallel to the axes of both cylinders. Notice that for every such cross-section, we always get a circle, which is from the sphere, inscribed in a square, which comes from the area of intersection of the two cylinders."



"Note that the ratio of the area of the circle to the area of the square at any cross-sectional level is a constant, namely:

$$\frac{\text{Area of Circle}}{\text{Area of Square}} = \frac{\pi r^2}{(2r)^2} = \frac{\pi}{4}$$

Since this is true for every cross section, we can say that the volume of the inscribed sphere and the volume of intersection of the two cylinders must be in the same ratio:

$$\frac{\text{Volume of Sphere}}{\text{Volume of Intersection}} = \frac{\text{Area of Circle}}{\text{Area of Square}}$$

$$\frac{\text{Volume of Sphere}}{\text{Volume of Intersection}} = \frac{\pi}{4}$$

$$\frac{\frac{4}{3}\pi r^3}{\text{Volume of Intersection}} = \frac{\pi}{4}$$

$$\text{Volume of Intersection} = \frac{16}{3}r^3$$

Interesting, eh, Isaac?"

"Thoroughly fascinating, Doctor."

"Well, then, how about this problem. Archimedes couldn't understand my solution quite that well, probably since my Greek was so horrible. Anyway, try finding the **total** volume of **three** cylinders of radius  $R$  and 'sufficient' length  $L$  whose axes of symmetry intersect at right angles."

"Well, Doctor, since we're looking for **total** volume of the three intersecting cylinders, I suppose we could use the Inclusion-Exclusion Principle to give us the formula:

$$\begin{aligned} A \cup B \cup C &= (A + B + C) - (A \cap B + B \cap C + A \cap C) + (A \cap B \cap C) \\ \text{Total Volume} &= 3 (\text{Volume of Cylinder}) \\ &\quad - 3 (\text{Volume of Intersection of Two Cylinders}) \\ &\quad + (\text{Volume of Intersection of Three Cylinders}) \end{aligned}$$

But finding the volume of intersection of **three** cylinders is going to be quite tough, isn't it, Doctor?"

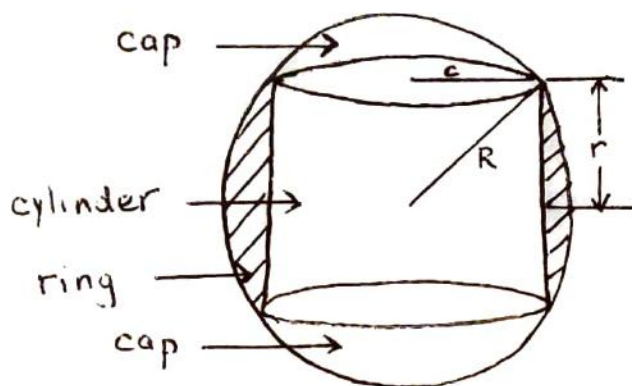
"Quite right, Ike. And if you try Calculus, you'll get stuck with quite a difficult integral to evaluate, with trigonometric substitutions I haven't even taught you yet. Try it."

After a few hours, Newton finally gave up on the problem. Using Calculus just didn't help. "Doctor," he said, "I'm absolutely lost on this problem. Can you show me the solution?"

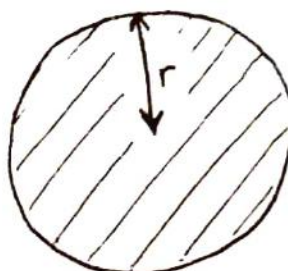
"Certainly, but first, we must prove a little theorem about rings."

"About what?"

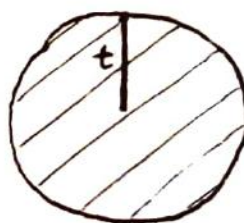
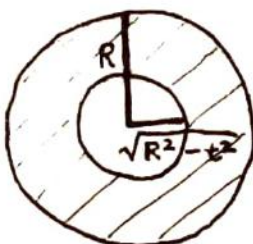
"You know, rings! Little circular objects. The type of ring we'll have to study is the ring that you get when you drill a hole of radius  $r$  through a sphere down its diameter. Compare the cross-sections taken perpendicular to the drillhole of such a sphere to the cross-sections of this smaller sphere:



Large Sphere's Ring:  
 $\pi R^2 - \pi(R^2 - t^2) = \pi t^2$



Small Sphere:  
 $\pi r^2$



So the cross-sections are equal, and therefore, so are the volumes  $\left(\frac{4}{3} \pi r^3\right)$ .

But we don't want the volume of the ring for our problem. We want the volume of the two 'caps,' you know, the spherical sections that we said were drilled out. To find that, we use simple subtraction:

$$\begin{aligned} 2 (\text{Volume of Caps}) &= (\text{Volume of Sphere}) \\ &\quad - (\text{Volume of Ring}) \\ &\quad - (\text{Volume of Cylinder}) \\ &= \frac{4}{3} \pi R^3 - \frac{4}{3} \pi r^3 - \pi c^2 2r \end{aligned}$$

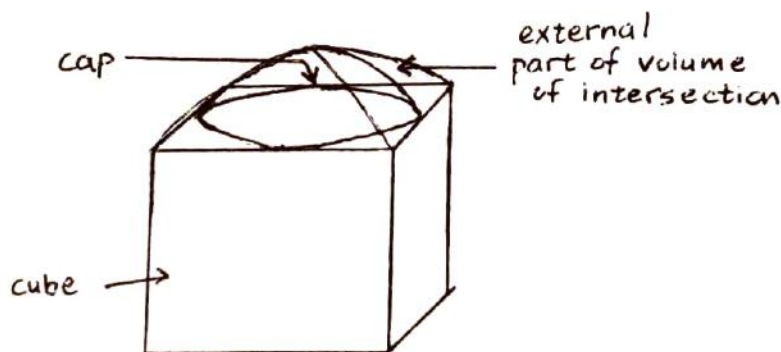
where  $c$  is the radius of the cylinder. Neat, isn't it?"

"Yes, it's all nice and pretty Doctor, but what's this for? We were talking about the intersection of three cylinders."

"All in good time. Remember that the intersection of our three cylinders is symmetric with respect to the three orthogonal axes, right? Well, let's put a sphere of radius  $R$  inside of this intersection. We can also place a cube of side  $R\sqrt{2}$  inside of the volume of intersection. The cube's volume is therefore  $2\sqrt{2} R^3$ . Now recall that this cube, since it can fit in any one of the cylinders, is entirely in the intersection of the three cylinders. Now we ask ourselves what else is included in this

intersection aside from this cube. Of course, if we study a single face of the cube, we note that the volume of intersection of the three cylinders which is adjacent to the face is entirely contained in one of the cylinders. Thus, we must only worry about the intersection of the remaining two cylinders. Again, we take cross-sections of the volume of intersection outside of the cube, and if we look at these cross sections, we get the same identical figures as we did for the intersection of two cylinders."

"Ah, then Doctor, the problem lies in that this time, we have no complete sphere with which to compare the volume of intersection to..."



"Which is precisely why we found the volume of the 'caps'! We can use the volume of these very 'caps' in the ratio just as we used the volume of the sphere in the two-cylinder intersection. But remember that we have 6 of these 'cap-derived' volumes. Thus,

$$6 (\text{Cap-Derived Volume}) = 6 (\text{Volume of Cap}) \times \frac{4}{\pi}$$

because, as in the case of the sphere and the intersection of two cylinders, the cross-sectional areas are in the constant ratio of  $\frac{4}{\pi}$ ."

"But we have to compute the volume of the cap first, Doctor. So I guess we have to use the fact that the volume of a pair of such caps is

$$\left[ \frac{4}{3} \pi R^3 - \frac{4}{3} \pi r^3 - \pi c^2 (2r) \right]."$$

"Right, Isaac. Well, you can easily see that  $R$  in our formula is simply the radius of our sphere inscribed in the three cylinders and that  $r = \frac{1}{2} (\text{Side of Cube}) = \frac{\sqrt{2}}{2} R$ . Also,  $c = r = \frac{\sqrt{2}}{2} R$ , since these are all measures of one-half the length of side of the cube. Substituting these facts in our cap equation, we get:

$$2 (\text{Volume of Caps}) = \frac{4}{3} \pi R^3 - \frac{4}{3} \pi r^3 - \pi (c^2)(2r)$$

$$\begin{aligned}
&= \frac{4}{3} \pi \left( R^3 - \frac{\sqrt{2}}{4} R^3 \right) - \pi \left( \frac{1}{2} R^2 \right) (\sqrt{2} R) \\
&= \pi R^3 \left( \frac{8 - 5\sqrt{2}}{6} \right)
\end{aligned}$$

And then

$$6 \text{ (Volume of Caps)} = \pi R^3 \left( \frac{8 - 5\sqrt{2}}{2} \right)$$

We then multiply this by  $\frac{4}{\pi}$ , which is our volume of intersection's cap to spherical cap ratio, to give us:

$$6 \text{ (Cap-Derived Volume)} = 4R^3 \left( \frac{8 - 5\sqrt{2}}{2} \right) = (16 - 8\sqrt{2}) R^3$$

Thus, we've found the volume of the intersection of the three cylinders which is outside of the inscribed cube. To find the total volume, we add the separate volumes:

$$\begin{aligned}
\text{(Volume of Intersection of 3 Cylinders)} &= \text{(Volume of Cube)} + 6 \text{ (Cap-Derived Volume)} \\
&= 2\sqrt{2} R^3 + (16 - 10\sqrt{2}) R^3 \\
&= (16 - 8\sqrt{2}) R^3
\end{aligned}$$

Do you get it, Isaac?"

"Now I see! We then go back to our Inclusion-Exclusion to find the total volume of this figure:

$$\begin{aligned}
\text{(Total Volume)} &= 3 \text{ (Volume of Cylinder)} - 3 \text{ (Volume of Intersection of 2 Cylinders)} \\
&\quad + \text{(Volume of Intersection of 3 Cylinders)} \\
&= 3 (\pi R^2 L) - 3 \left( \frac{16}{3} R^3 \right) + (16 - 8\sqrt{2}) R^3 \\
&= 3\pi R^2 L - 8\sqrt{2} R^3
\end{aligned}$$

That's wonderful, Doctor! Doctor?"

Isaac Newton looked all around, and that mysterious friend of his, the Doctor, had disappeared. Little did poor Isaac know that the Doctor was already on his way to visit Mr. Leibnitz's.

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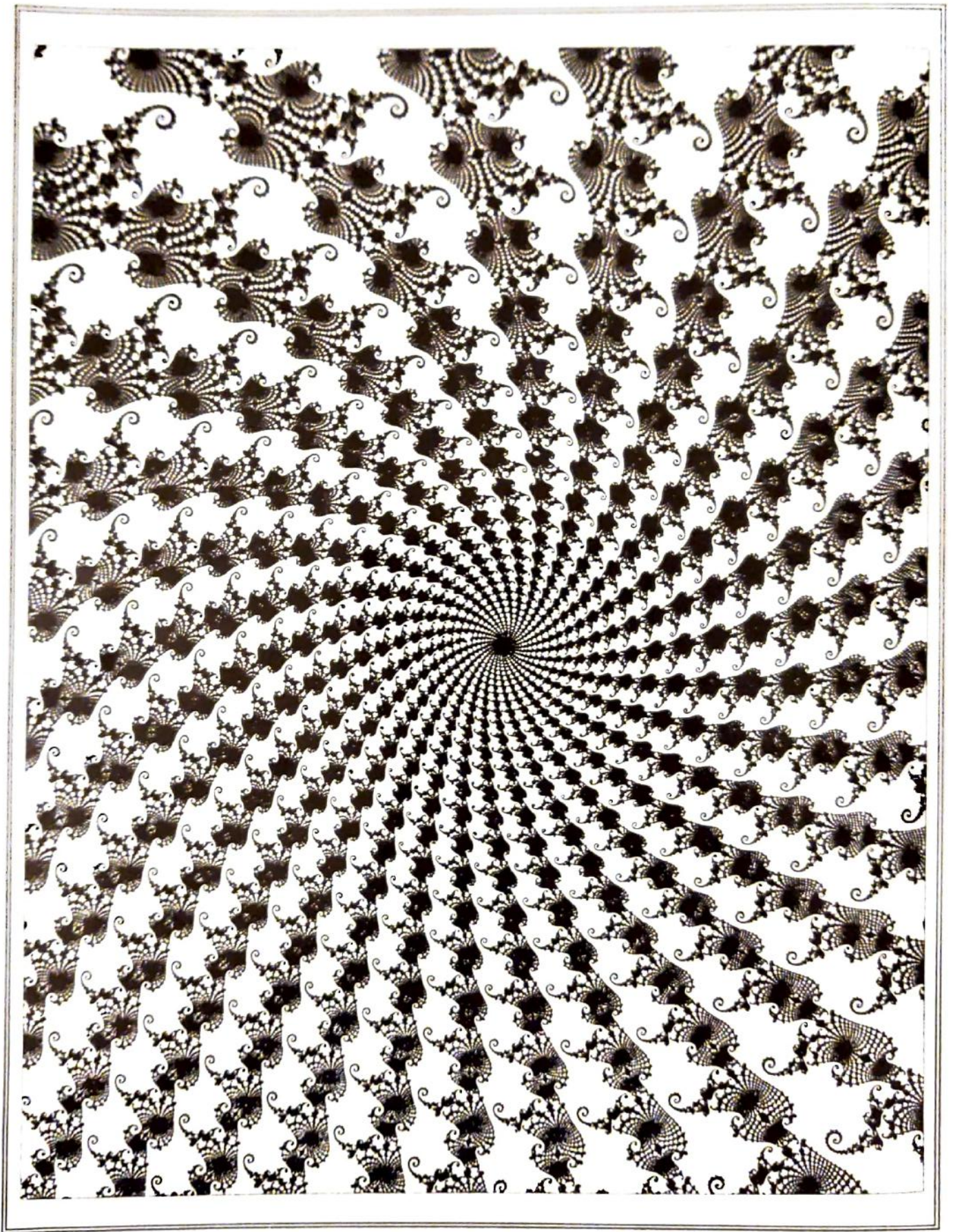
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